The Great Chai Stall Crawl

Meera is a food blogger visiting the old-city market, which is famous for its chai. The market has `n` chai stalls, numbered `1` to `n`, connected by `m` one-way pedestrian lanes (the lanes are one-way to keep the crowds moving). Walking a lane means paying a small rickshaw-style fare, and every lane has its own fare.

Meera's plan for the day is strict. She enters the market at stall `1` and must leave at stall `n`, where the metro station is. In between, she wants to taste chai at **every stall exactly once** — no stall skipped, no stall visited twice. Stall `1` is the first stall she tastes and stall `n` is the last.

Such a plan is called a **crawl**. Two crawls are different if the sequences of lanes they use are different. This matters when the market has two separate lanes between the same pair of stalls: they are different lanes, so using one or the other gives two different crawls.

Meera's blog post needs two numbers:

1. The number of different crawls, modulo `1,000,000,007`.
2. The cheapest total fare among all crawls.

If no crawl exists, report `0 -1`.

Examples

Example 1

Input:

```
4 6
1 2 3
1 3 1
2 3 2
3 2 1
2 4 4
3 4 6
```

Output:

```
2 6
```

Explanation: Two crawls visit every stall exactly once. `1 -> 2 -> 3 -> 4` *costs 3 + 2 + 6 = 11, and* `1 -> 3 -> 2 -> 4` *costs 1 + 1 + 4 = 6. So there are 2 crawls and the cheapest costs 6.*

Example 2

Input:

```
3 2
1 2 5
3 2 4
```

Output:

```
0 -1
```

Explanation: There is no lane out of stall 2, so after tasting stalls 1 and 2 Meera can never reach stall 3. No crawl exists.

Example 3

Input:

```
3 3
1 2 5
1 2 3
2 3 4
```

Output:

```
2 7
```

Explanation: The route `1 -> 2 -> 3` can use either of the two lanes from stall 1 to stall 2, so there are 2 different crawls. Their fares are 5 + 4 = 9 and 3 + 4 = 7, so the cheapest is 7.

Input Format

The first line contains two integers `n` and `m` — the number of stalls and the number of one-way lanes.

The next `m` lines each contain three integers `a`, `b`, and `fare` — a one-way lane from stall `a` to stall `b` that costs `fare` to walk.

Stalls are numbered from `1` to `n`.

Constraints

- `1 ≤ n ≤ 20`
- `0 ≤ m ≤ n²`
- `1 ≤ a, b ≤ n`
- `1 ≤ fare ≤ 1,000,000`
- Self-lanes (`a = b`) and duplicate lanes between the same pair of stalls may occur.

Output Format

Print two space-separated integers: the number of different crawls modulo `1,000,000,007`, and the cheapest total fare among all crawls.

If there is no crawl, print `0 -1`.

If `n = 1`, Meera only needs to taste stall `1` and does not walk any lane, so print `1 0`.

Sample Input 0

```
4 6
1 2 3
1 3 1
2 3 2
3 2 1
2 4 4
3 4 6
```

Sample Output 0

```
2 6
```